SUBHANSHU SETHI

sethisubhanshu@gmail.com | github.com/subhanshusethi | linkedin.com/in/subhanshusethi | +91 9958777303

WORK EXPERIENCE

Computer Vision Engineer (AI/ML) — **Novus Hi-Tech, Gurugram**
Jul 2025 – Present

- **Developing** a dual-arm wire quality inspection system on **Anvil's OpenArm** and **SO101** robots using **imitation learning** policies (**ACT, SmolVLA, Groot**); conducting empirical evaluation of **VLA architectures** for **language-conditioned, multi-step** fine-grained manipulation.
- **Deployed** real-time RGB-D pose estimation pipeline in **C++/ROS**, achieving **25–30 Hz** inference via **OpenVINO** for heterogeneous execution (**CPU, iGPU, NPU**) on Intel NUC Ultra.
- **Engineered** a Reflector-Based Localization (RBL) system integrating **LiDAR** temporal clustering and triangulation, achieving a pose accuracy of **5–10 cm** and boosting navigation resilience in repetitive environments.
- **Optimized** pallet detection by migrating from **EfficientDet** to **YOLOv8** via **TensorRT** on the resource-constrained Jetson Nano, achieving a **3x throughput increase** (from **5–7 FPS** to **15–20 FPS**).
- **Automated** complete ML training and versioning workflows using **DVC** and agentic coding principles, significantly accelerating offline annotation and model staging.

Computer Vision Intern — **Novus Hi-Tech, Gurugram**
Jun 2024 – Jun 2025

- **Benchmarked** real-time models (**YOLO, SAM**) and **optimized** RGB-D Pose Estimation by eliminating **RANSAC**, gaining proficiency across the full edge-robotic stack (**Controls, Navigation, Perception**).

EDUCATION

B.Tech in Electrical Engineering — **Delhi Technological University**
Courses: Deep Learning, Computer Vision, Computer Networks, Cloud Computing, IoT — *2021 – 2025*

PUBLICATIONS

- **GExSent: Gated Experts for Robust Sentiment Analysis Across Modalities** — IJCNN 2025, Rome, Italy
- **ContXCLIP: Contextual Attention for Vision-Language Understanding** — Under Review

PROJECTS

GExSent — Multimodal Classification | GitHub — *Feb 2025 – Apr 2025*
Tech Stack: PyTorch, CLIP, ModernBERT, Mixture of Experts

- Achieved **SOTA performance** on MOSI, MOSEI, and Memotion benchmarks (**MAE 0.865** and **F1 87.24**) with a highly parameter-efficient **9M trainable pipeline** (Accepted at **IJCNN 2025**)
- Architected a Hierarchical Gated MoE and a Gated Attention Mechanism to robustly fuse multi-modal sentiment logic.

ContXCLIP — Image Captioning | GitHub — *Jul 2024 – Feb 2025*
Tech Stack: PyTorch, Vision-Language Models (CLIP), GPT-2

- Outperformed prior baselines on MS-COCO datasets, securing **CIDEr 134.1** and **BLEU-4 39.4** metrics with an **11.7M parameter** lightweight architecture.
- Designed a context preservation module (CPM) and cross-modal contextual attention (XCA) to align visual and text domains.

GoldenAid — Bystander First Aid AI | GitHub — *Apr – May 2026*
Tech Stack: Android, Gemma 4 E2B, LiteRT-LM, YOLO11n, MediaPipe BlazePose, Unsloth

- **Built** a fully on-device Android app combining YOLO11n person detection, MediaPipe BlazePose pose estimation, and Gemma 4 E2B multimodal inference via LiteRT-LM — achieving 3.2s end-to-end latency with zero internet dependency.
- **Fine-tuned** Gemma 4 E2B using Unsloth + QLoRA on 3,000 first aid scenarios, producing action-first bystander-appropriate responses; integrated GPS incident logger and one-tap 108 emergency dialer.

ROS Agentic Debugger — *Mar 2026 – Apr 2026*
Tech Stack: ROS, Python, LLMs (Gemma 4, Ollama)

- Automated live robotics validation by leveraging **Gemma's native tool-calling capabilities** to deeply reason over ROS topics, TF trees, and Hz node flows.
- Eliminated manual pipeline checks, enabling instantaneous diagnostic health verification aboard autonomous platforms.

ACHIEVEMENTS

- **Top 15 out of 19,000+ Teams** (Flipkart Grid Jan '25): **Developed** an FMCG feature extraction pipeline using **DINOv2** and **Qwen2-VL**, **fine-tuned with PEFT (LoRA/QLoRA)**, for zero-shot detection and feature extraction.
- **6th Place Globally** (ICUAS Mar'25): **Developed** a multi-drone navigation system employing ArUco marker vision, BFS/A* routing, and spatial raycasting for obstacle avoidance.
- **3rd Place** (IMAV Nov'23): **Built** an autonomous lane-following drone infused with classical CV and PID control.

TECHNICAL SKILLS

Languages: Python, C++
ML / CV: PyTorch, TensorFlow, OpenCV, Transformers, LoRA, CLIP, YOLO, TensorRT, ONNX Runtime, OpenVINO
MLOps: Docker, CI/CD (Git, DVC), MLflow, W&B, FastAPI (REST), AWS (Lambda, EC2, S3, ECS)
Robotics: ROS, SLAM, Imitation Learning, VLA, Jetson Nano, Intel NUC Ultra, RealSense D415, Raspberry Pi
Domains: Computer Vision, Deep Learning, Embodied AI, Model Optimization, Machine Learning, Agentic AI